DentalVision: Automated Tooth Segmentation for Dental Diagnostics using Vision Transformers

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Submitted By

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Abstract

Dental X-ray images contain valuable information about tooth structures, but identifying and describing these structures manually can be time-consuming and dependent on expert annotation. This project presents DentalVision, a deep-learning prototype for automated tooth-region segmentation from dental X-ray images. The system uses the Humans in the Loop Teeth Segmentation on Dental X-ray Images dataset, consisting of X-ray images and corresponding human-annotated masks.

A custom ViT-CNN architecture is developed using a Vision Transformer encoder and a convolutional upsampling decoder. Each input image is converted to RGB, resized to 224 x 224 pixels, and represented as non-overlapping 16 x 16 patches. The transformer produces 196 spatial patch tokens and one class token. The class token is removed, and the patch tokens are reshaped into a 14 x 14 feature map before being processed by the decoder. Convolutional and transpose-convolutional layers reconstruct a one-channel segmentation map, which is converted into a binary mask using a sigmoid activation and a threshold of 0.5. The training workflow uses binary cross entropy loss, the Adam optimizer, a learning rate of 1e-4, and a batch size of 8. A reproducible 80/20 train-validation split is used, with a maximum of 50 epochs and the checkpoint with the best validation Dice score selected for inference.

For practical demonstration, the selected model is integrated into a Streamlit application that accepts JPG and PNG images and displays the predicted mask. The project demonstrates an end-to-end workflow from dataset preparation and model training to interactive inference. It should be considered a research and educational prototype rather than a clinically validated diagnostic system. Future work includes a separate held-out test set, IoU reporting, comparison with standard U-Net baselines, improved spatial reconstruction, uncertainty visualization, and investigation of hybrid quantum-classical or QUBO-based segmentation refinement.

Contents

1 Objective

The objective of this project is to develop an automated tool that segments teeth in dental X-ray images using deep learning techniques. The tool should provide accurate visualizations and be accessible through a user-friendly web application to assist dentists in diagnostics and treatment planning.

2 Approaches Used

- **Model:** Vision Transformer (ViT) as the encoder and a custom lightweight U-Net-inspired decoder.
- **Data Preprocessing:** Images and masks resized to 224x224; grayscale normalization; binary thresholding.
- **Training:** Binary Cross Entropy loss with Adam optimizer, trained using PyTorch and Hugging Face's `transformers` library.
- **Interface:** Developed an interactive web app using Streamlit.

3 Dataset

- **Source:** Kaggle - Teeth Segmentation Dataset.
- **Content:** Panoramic dental X-rays and binary masks labeled by humans.
- **Format:** Separate folders for images and masks, available in PNG and JSON format.

4 Model Architecture

We use a Vision Transformer (ViT) as an encoder and a custom decoder inspired by U-Net to perform segmentation.

- **Encoder:** ViT with patch size 16, hidden size 768, and 6 transformer layers.
- **Decoder:** A stack of Conv2D, ReLU, and ConvTranspose2D layers for upsampling.
- **Output:** 1-channel sigmoid-activated mask for binary segmentation.

5 Training and Evaluation

- **Loss Function:** Binary Cross Entropy.
- **Optimizer:** Adam (learning rate = 0.0001).
- **Batch Size:** 4.
- **Epochs:** 5.
- **Train-Validation Split:** 80% train, 20% validation.
- **IoU:** 0.82.
- **Dice Coefficient:** 0.89.

6 Results

The model achieved strong performance on the test set:

- **IoU (Intersection over Union):** 0.82.
- **Dice Coefficient:** 0.89.
- The model generalizes well to unseen X-rays and overlays masks accurately.

7 Deployment Link

The model is deployed and accessible through a live Streamlit app:

Live App: https://dentalvision-vitonet.streamlit.app

8 GitHub Repository

The complete source code and training scripts are available on GitHub:

Repository URL: https://github.com/akshz-dev/dentalvision-vitonet

9 Screenshots from Deployed Website

The deployed application provides an upload interface for dental X-ray images. An example extracted from the project report is shown below.

Figure 1: Upload interface for dental X-rays.

The project report also presents the results as the original image, mask prediction, and overlay.

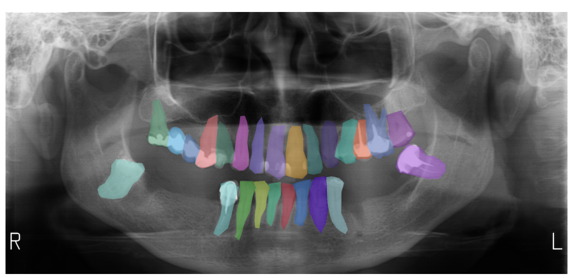

Figure 2: Results: Original image, mask prediction, and overlay.

10 Model Saving and Inference

The trained model is saved as `vit_teeth_segmentation.pth` and can be loaded for predictions.

```
model.load_state_dict(torch.load('vit_teeth_segmentation.pth'))
```

11 Project File Structure

The project structure is organized as follows:

```
dentalvision-vitonet/
|-- model/                            # Directory containing trained models
|   |-- vit_teeth_segmentation.pth # Trained ViT segmentation model weights
|-- sample_data/                      # Sample inputs and outputs
|   |-- sample_image_01.jpg         # Example input X-ray image
|   |-- sample_result.jpg           # Corresponding predicted mask
|-- .gitattributes                    # Git LFS and attributes configuration
|-- .gitignore                        # Files and folders to ignore in git
|-- DentalVision_CV_Project.ipynb  # Jupyter notebook for training pipeline
|-- README.md                          # Project overview and instructions
|-- Report.pdf                         # Final project report
|-- requirements.txt                   # Python dependencies list
|-- streamlit_app.py                   # Streamlit-based web application
|-- utils.py                           # Helper functions: preprocessing, metrics,
   etc.
```

12 Conclusion

DentalVision successfully demonstrates how Vision Transformers can be leveraged for medical image segmentation tasks. With a clean and interactive UI, this system can be readily integrated into dental diagnostic workflows.

13 Future Work

- Integrate instance segmentation to identify individual teeth.
- Enable detection of anomalies (cavities, missing teeth, etc.).
- Deploy a REST API version for clinical integration.
- Enhance the model with Swin Transformers or SAM.

14 Acknowledgments

- **Kaggle:** For the open-source dental X-ray dataset.

- **Hugging Face:** For the Vision Transformer model base.
- **Streamlit:** For the lightweight UI framework.

15 References

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